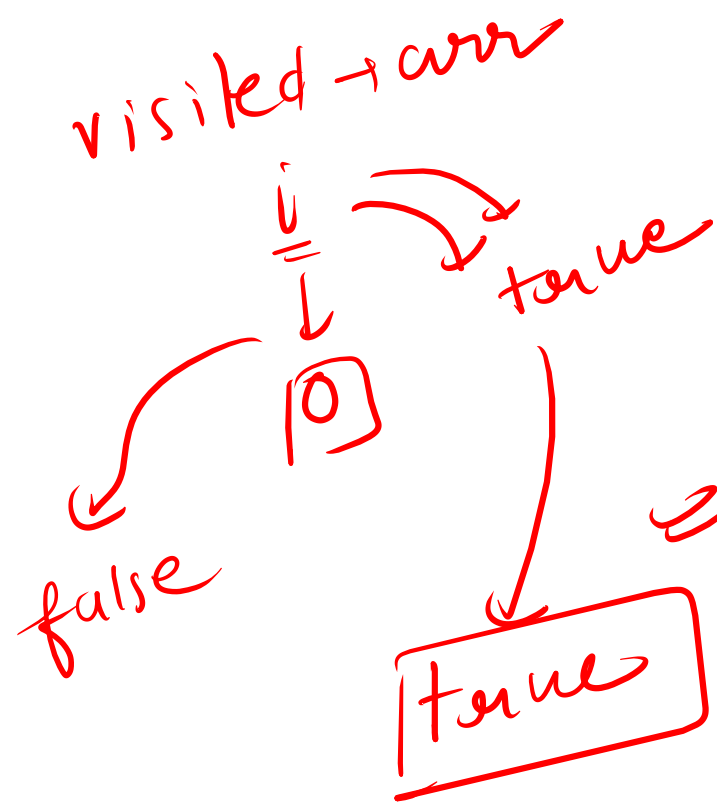
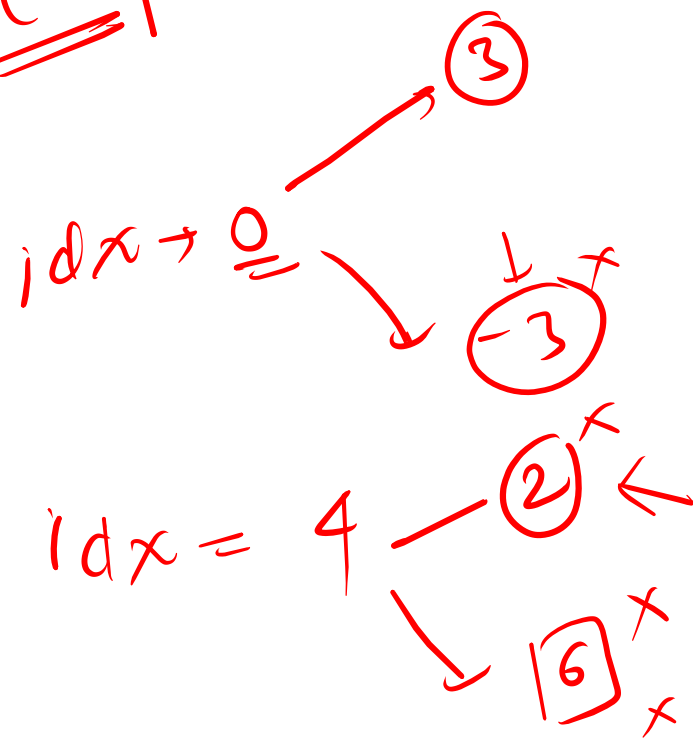
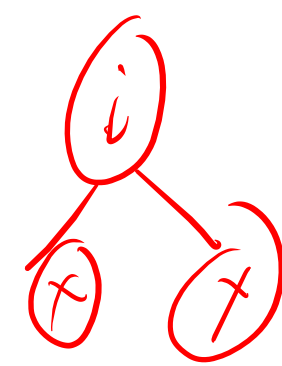
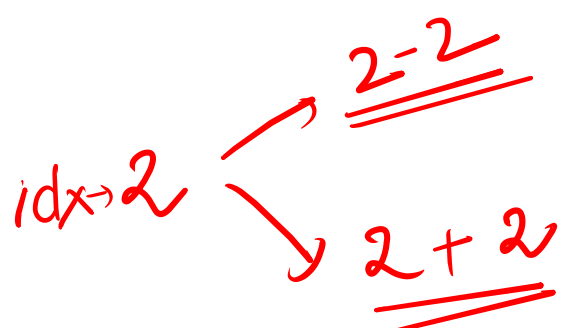
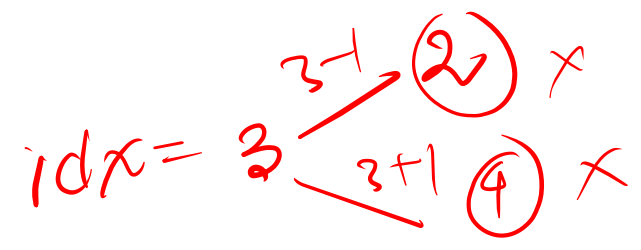
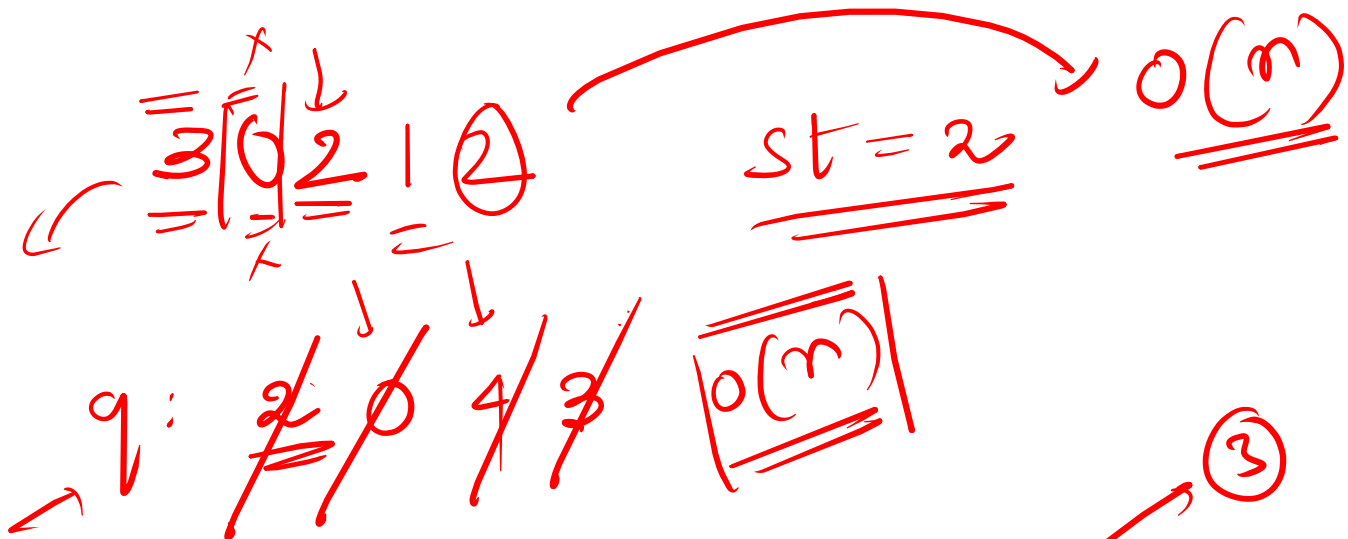


false



⇒ 1 2 → 5 smallest numbers

{
1
2
1 1
1 2
2 1
2 2
1 1 1
1 1 2
}

8

(K)

K=6

O(K)

num → num 1
 → num 2

1 → 1 1
 → 1 2

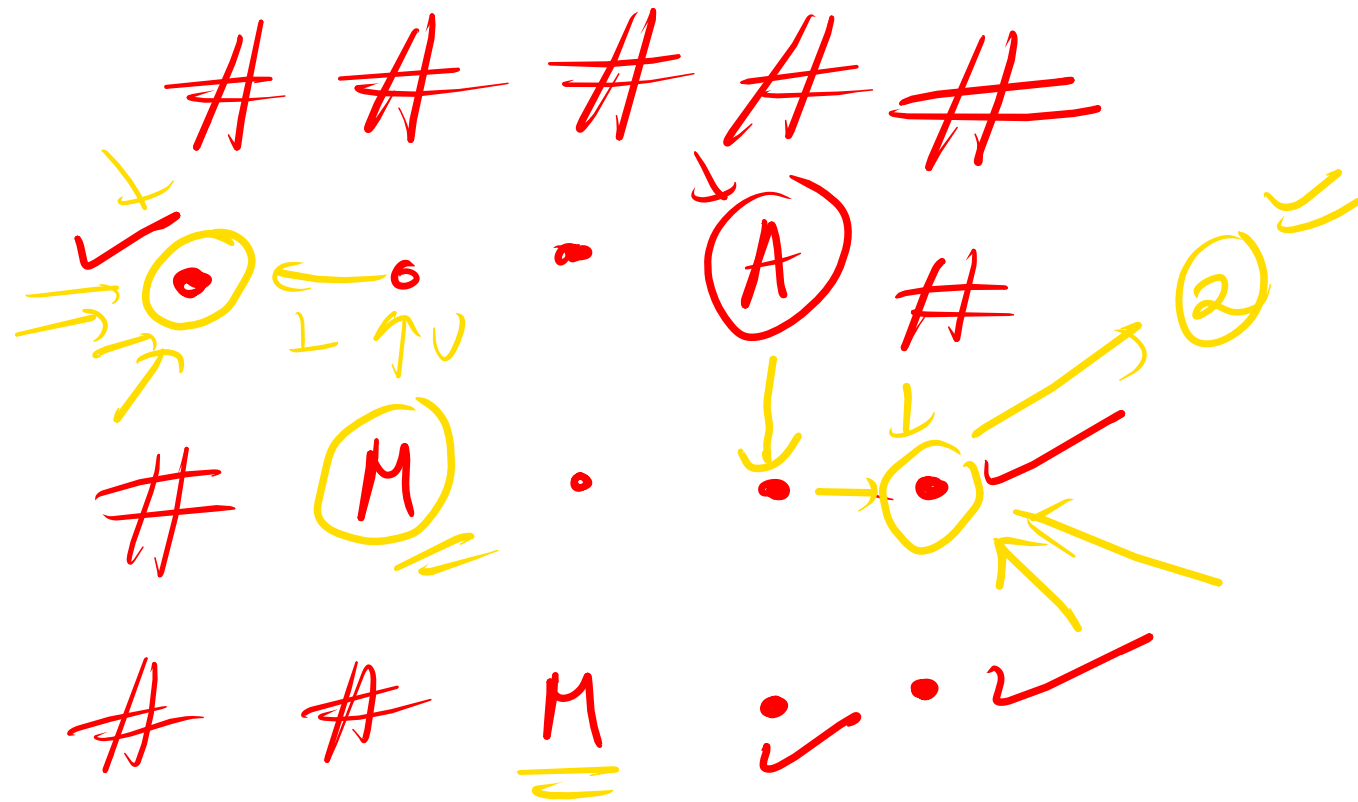
2 → 2 1
 → 2 2

count → K

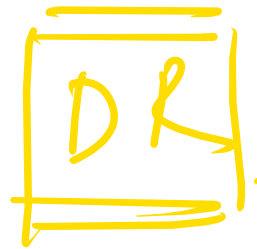
1 1 1 1 1 1
1 : ~~1~~ ~~2~~ ~~1 1~~ ~~1 2~~ 2 1
 2 2

1 1
1 2

2 1
2 2

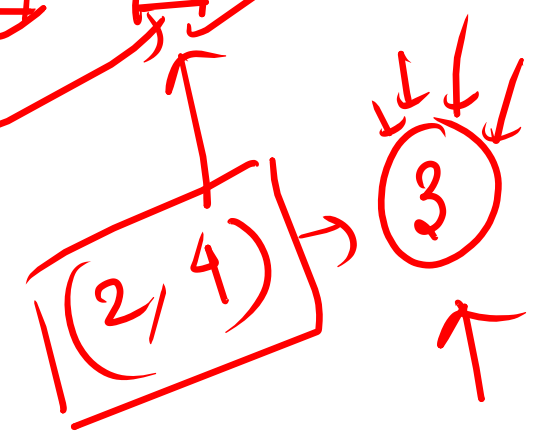


any
path





$\{M, M\} \rightarrow \underline{\underline{9}}$
 $(1, 0) \rightarrow 1$
 $(3, 3) \rightarrow 1$
 $(3, 4) \rightarrow 2$



BFS
 $A \rightarrow \{1, 3\} \{1, 2\} \{2, 3\}$
 $\downarrow$
 $\boxed{10} \quad 1 \quad 1$

$\{1, 1\} \rightarrow 2$
 $\{2, 2\} \rightarrow 2$
 $\{2, 4\} \rightarrow 2$
 $\{3, 3\} \rightarrow 2$
 $\{1, 0\} \rightarrow 3$
 $\{3, 4\} \rightarrow 3$
 $\rightarrow \boxed{\underline{\underline{DR}}}$
 $\textcircled{2}$

$\uparrow \text{mid} \rightarrow \text{ans}$

$T \rightarrow [i, j]$
 $\rightarrow f[i][j]$

max
time
I can
wait

0	5	#	#	F
1	2		F	
2	7		F	#
3	#		#	#
4	5			ε

4	#	2	#	0
3	2	1	0	1
3	2	1	0	#
4	#	2	#	6
5	4	3	4	5

10 109
 $0/0 \rightarrow \text{mid}$
 $\downarrow$
 $\text{mid} + \text{mid} + 1$
 $\downarrow$
 $\text{mid} + 2$

int

①
 $0 \dots \dots \dots \infty \rightarrow 10^9$
 $0 \dots \text{mid} \dots 10^9 \rightarrow \text{BS}$

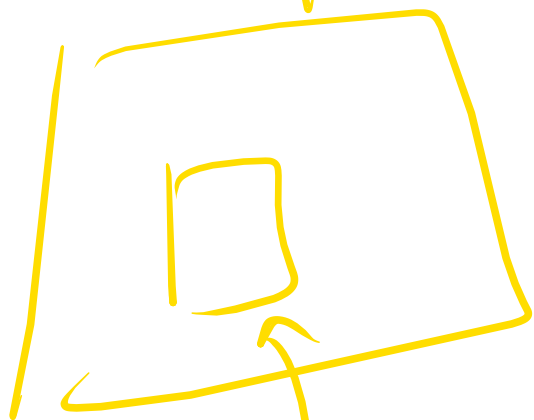
init $\rightarrow s \rightarrow \infty$
 $\downarrow x-1$
 $\rightarrow x-2 \dots \dots \dots 0$

10

mid
 $\sim [i, j]$
 $\rightarrow f[i][j]$

0 --- 10^9 $\leftarrow$ int

Graphs



Tree



unique path.

Tree

n → nodes

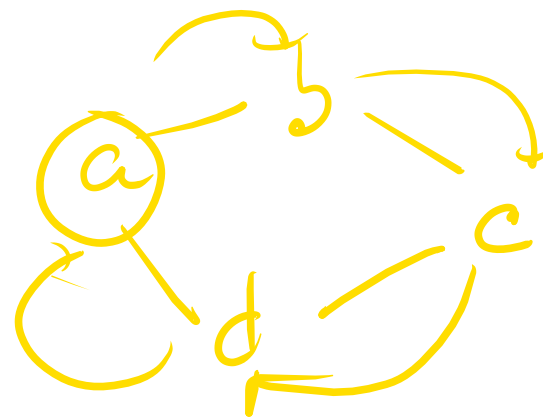
n-1 edges

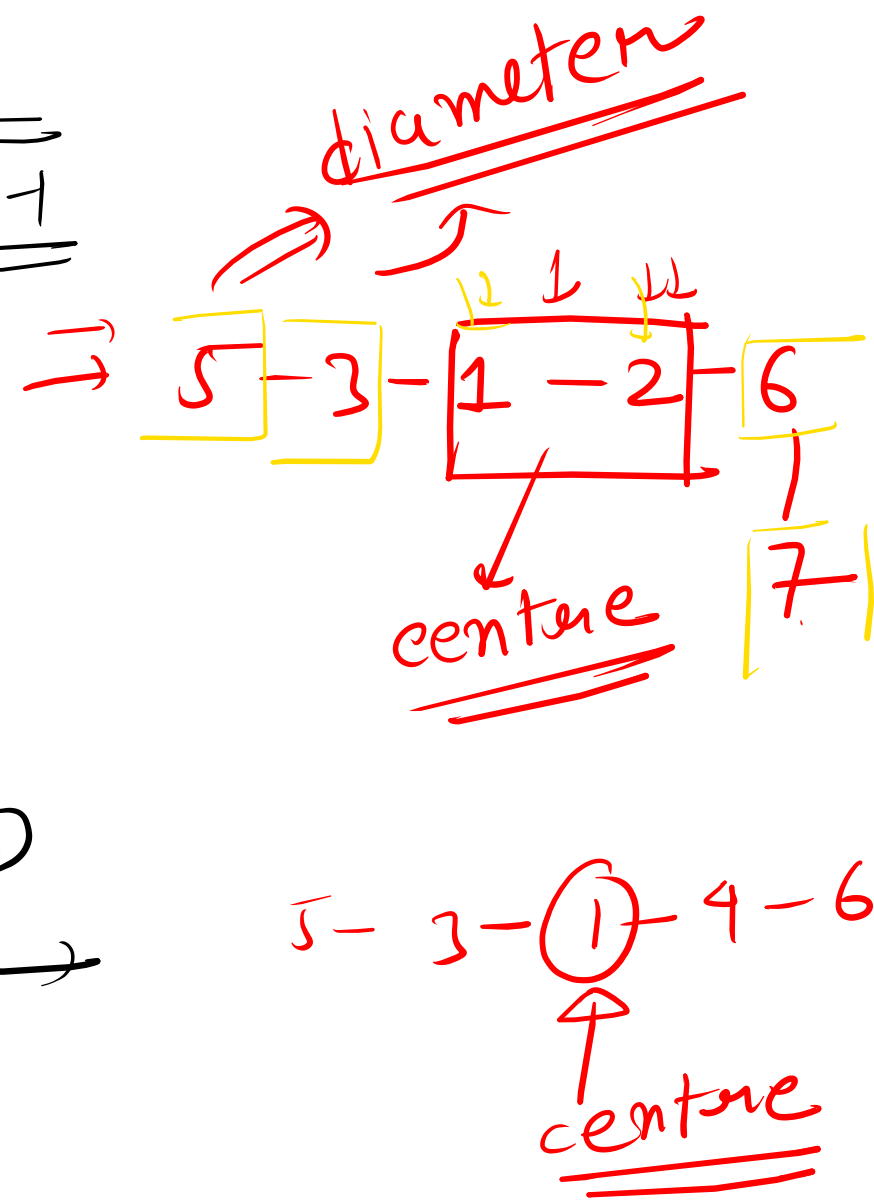
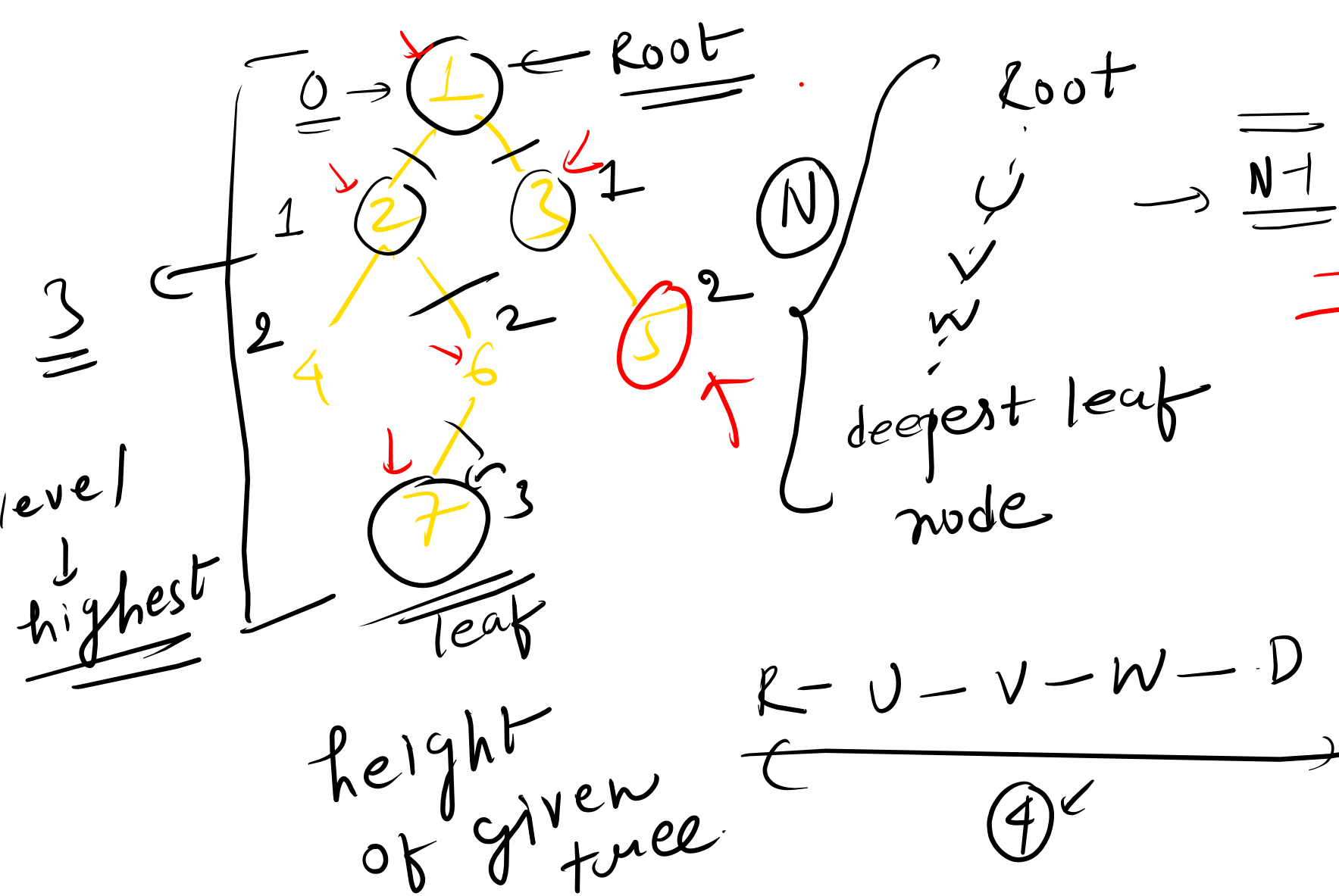


7 nodes

6 → edges.

cycle → doesn't exist





centre

$3: 2$
 $7: 1$
 $1: 3$
 $2: 3$
 1
 1

centre of tree

nodes

time:

dist to all
other nodes

to
minimised

$4 \rightarrow 1$

$7 \rightarrow 1$

$5 \rightarrow 1$

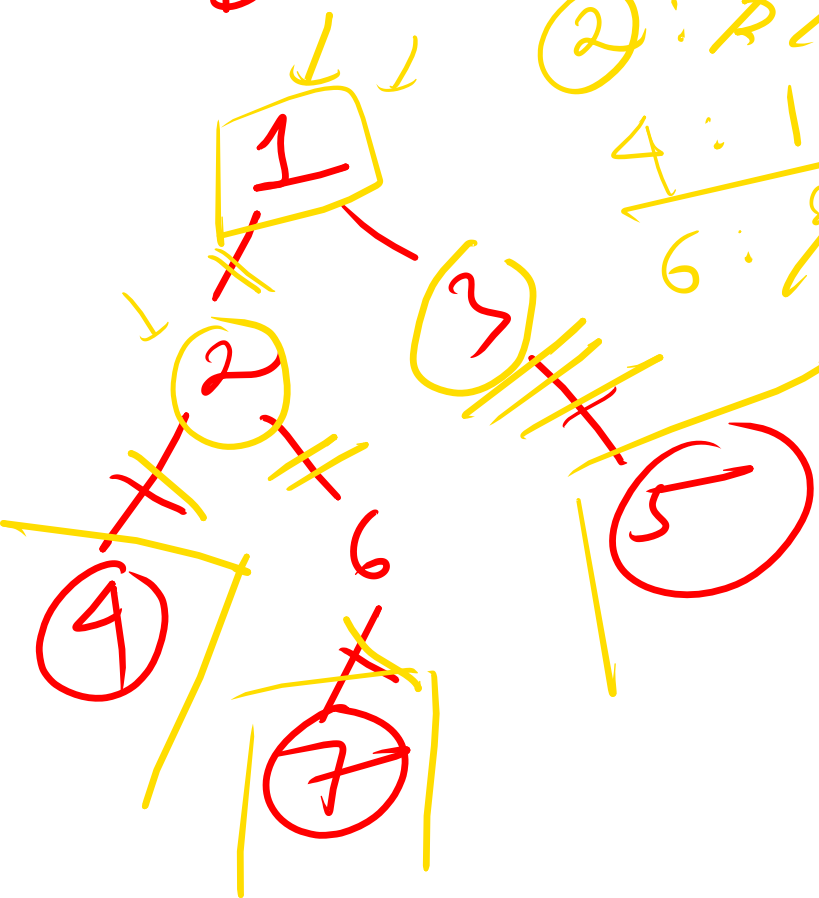
$6: 2$

$3: 2$

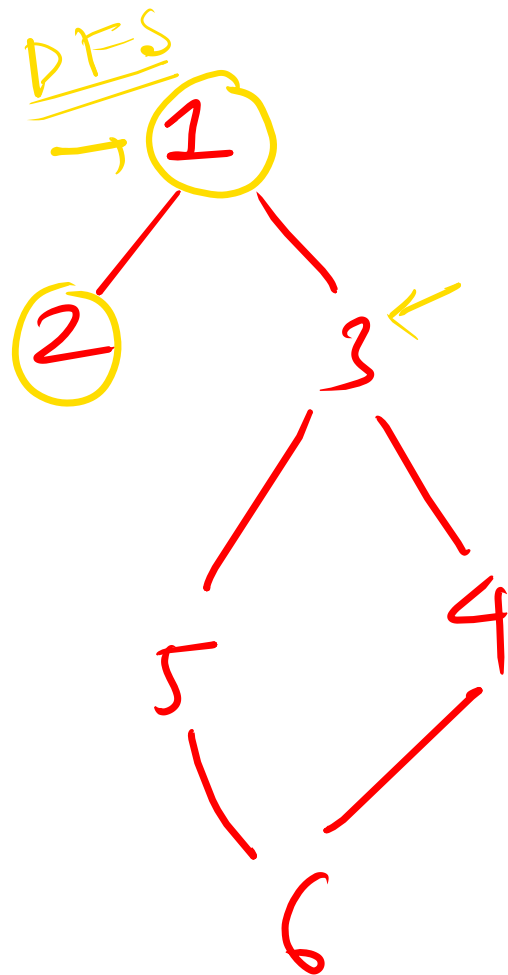
$9: 4$
 $7: 5$
 $6: 3$
 $2: 2$
 $1: 1$

centre

$2: 3$
 $1: 3$

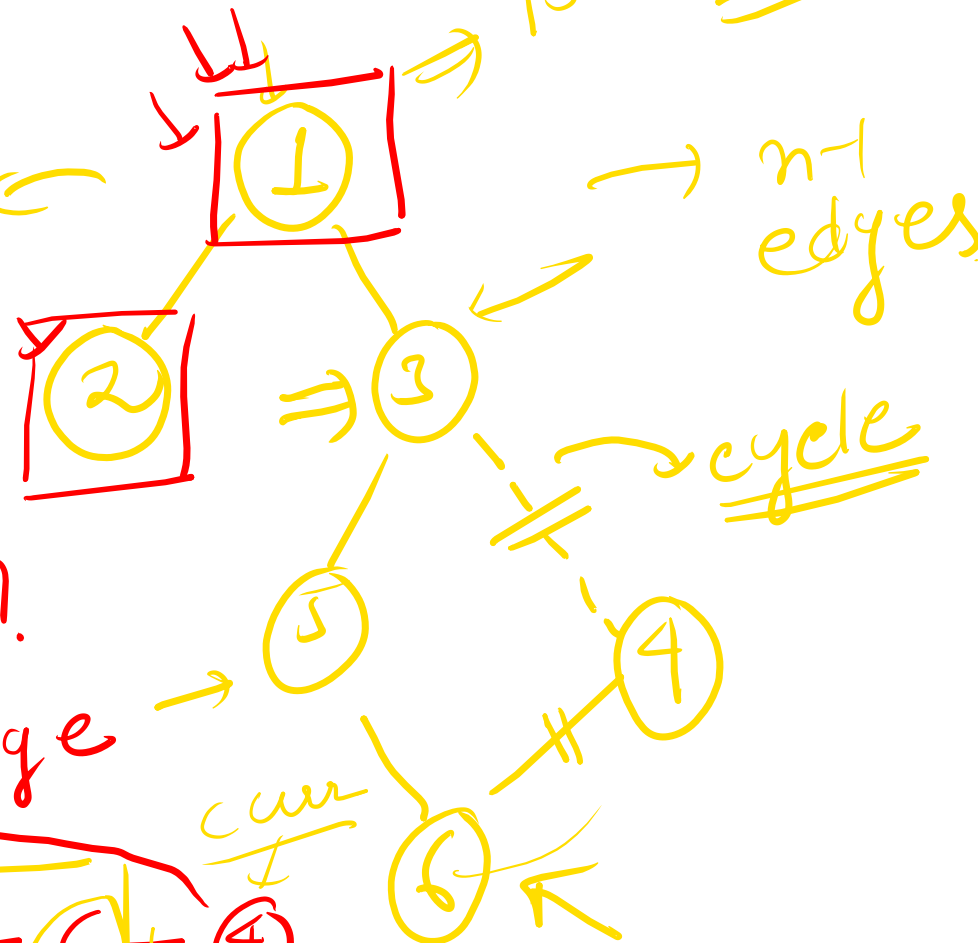


Cycle



cycle ??

DFS Tree

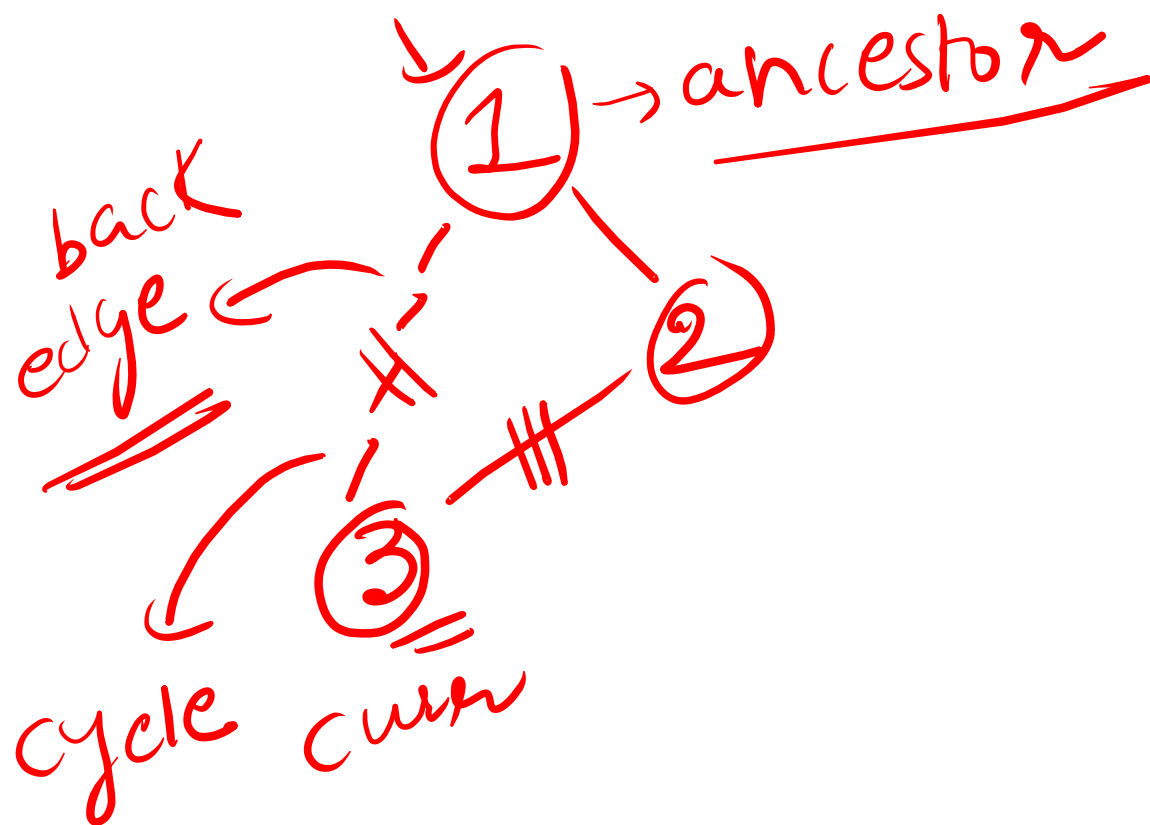
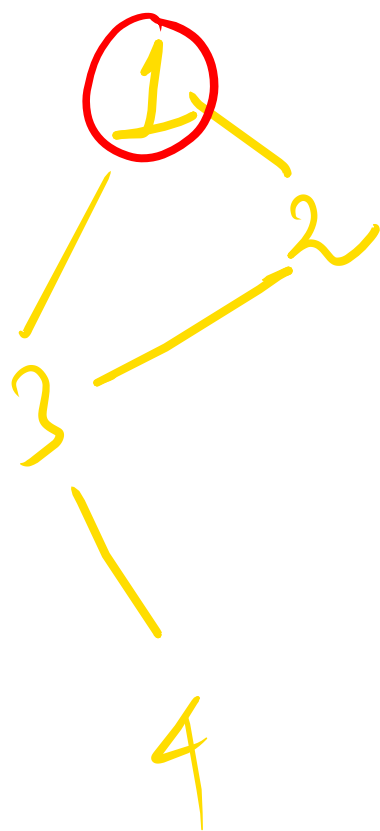


cycle ??

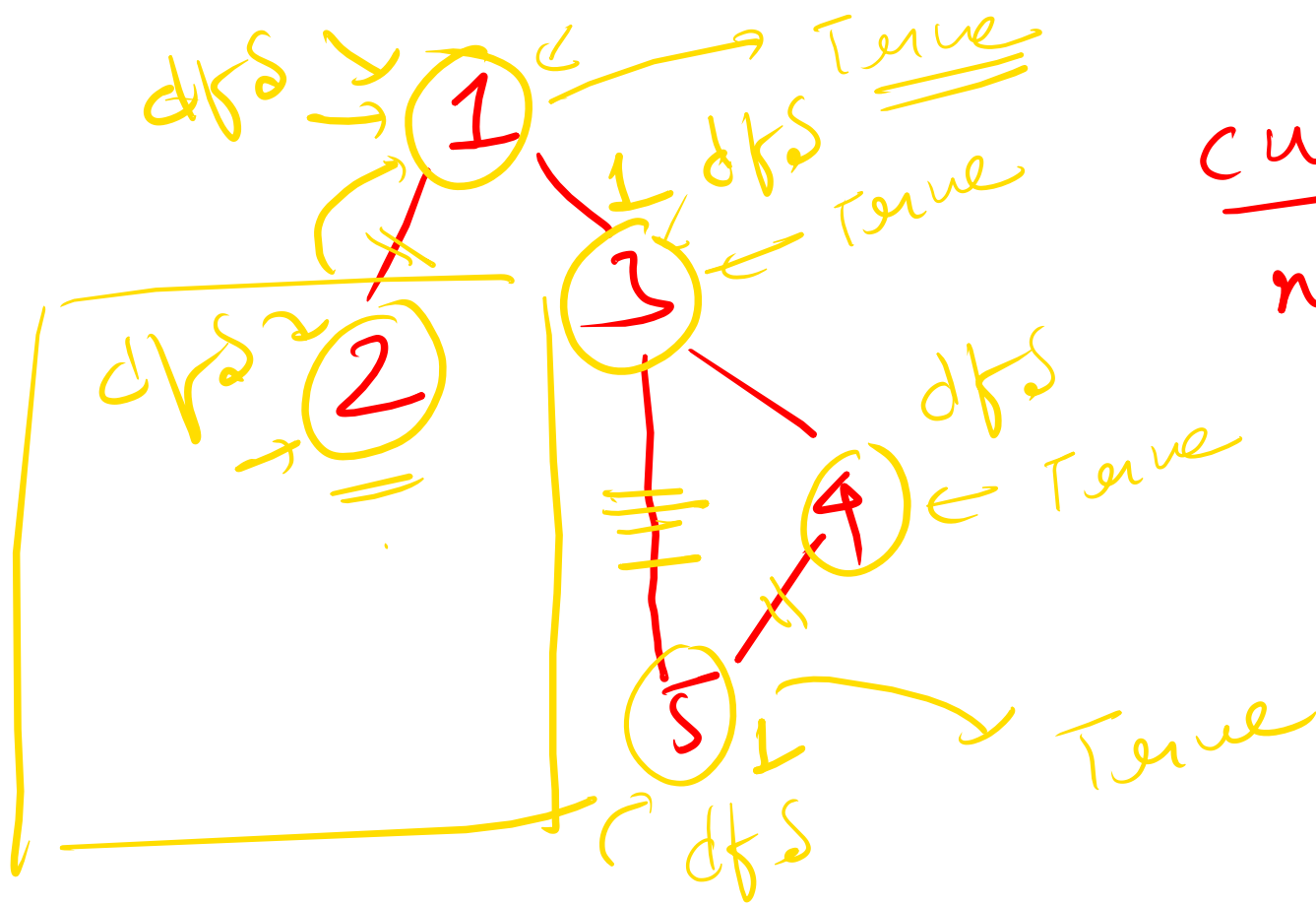
back edge

ancestor





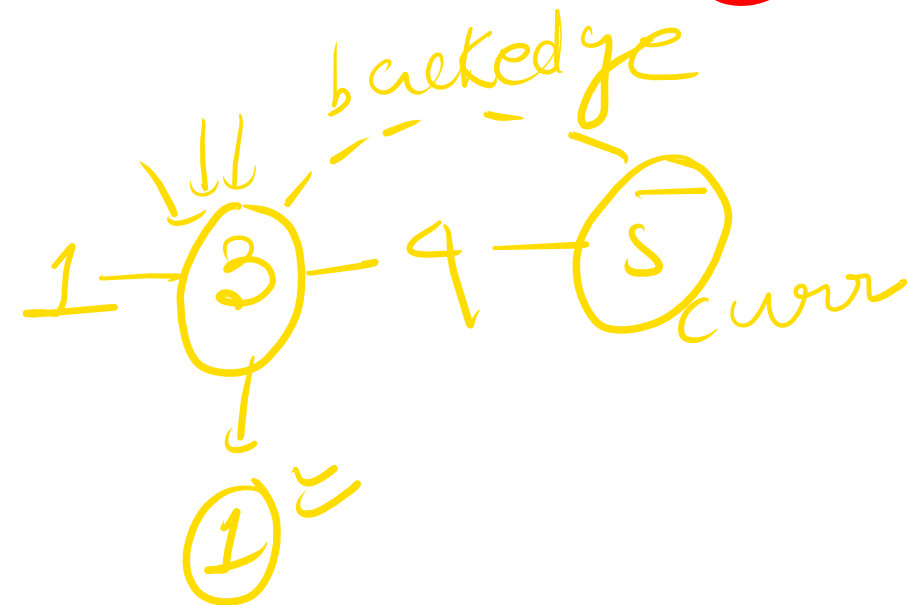
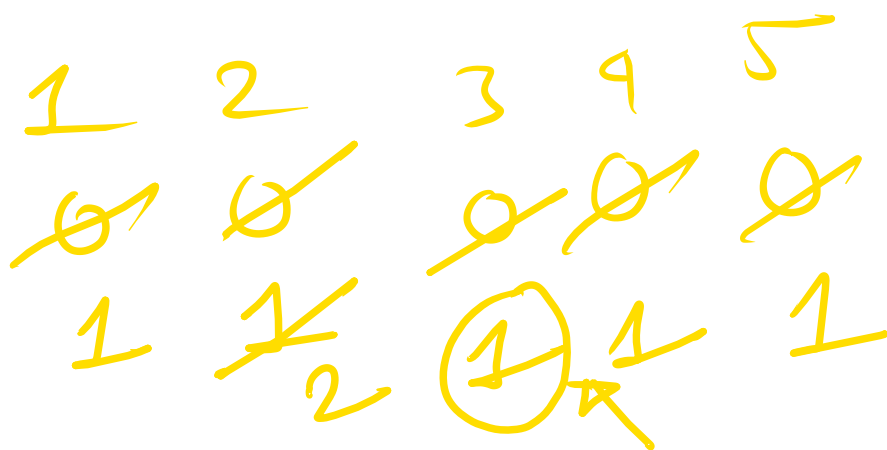
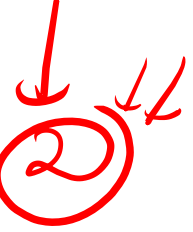
True/false

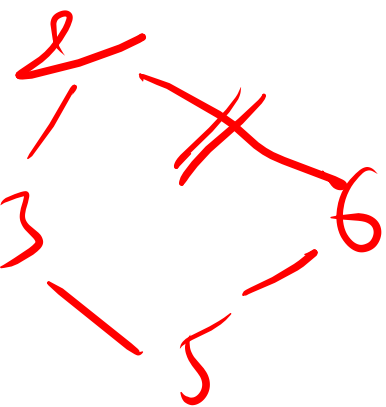
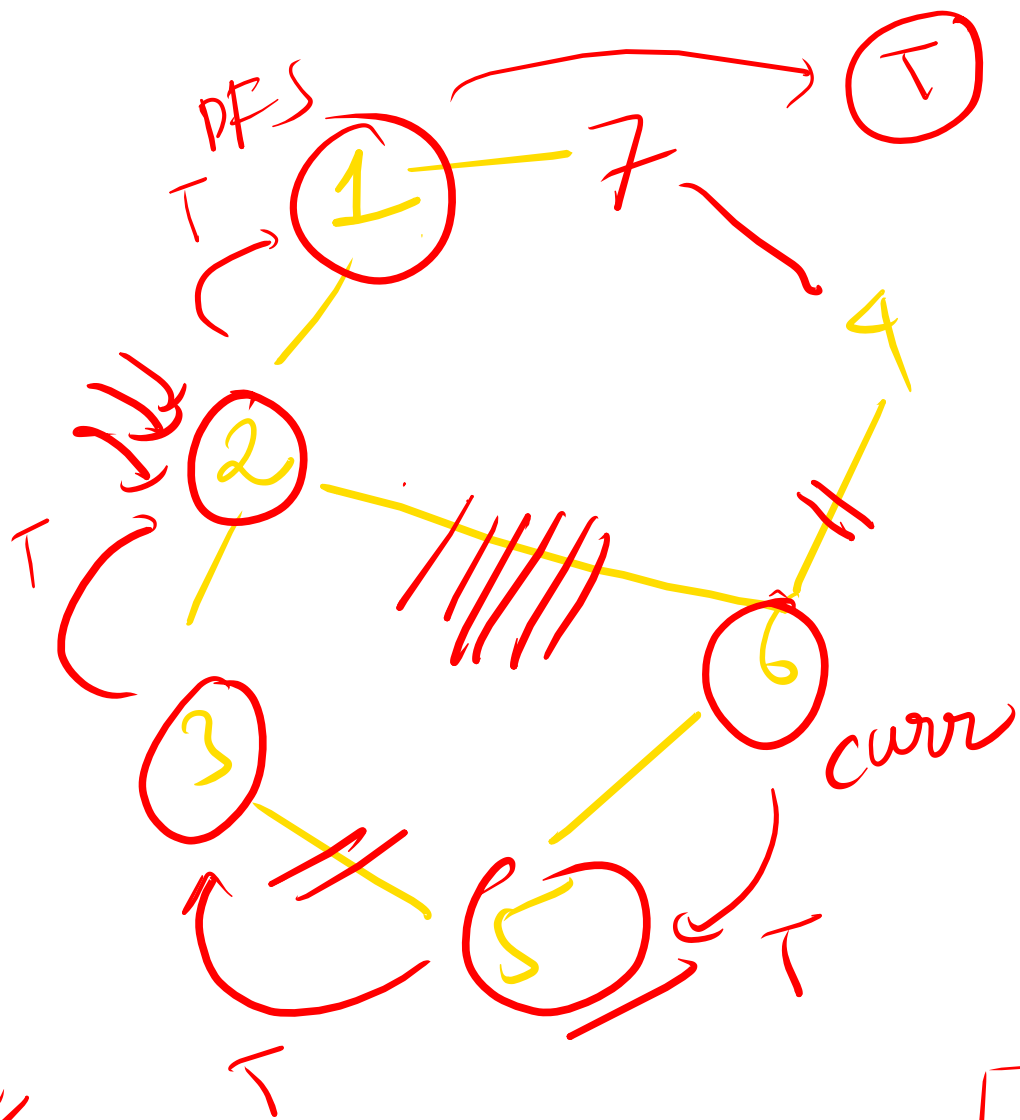


curr path → nodes → ①

node → unvisited → 0

nodes → visited → X path





$$\frac{\times}{\lfloor \frac{n}{2} \rfloor}$$

②

x

T/F

	1	2	3	4	5	6	7
color							
	1	2	3	4	5	6	7
	∅	∅	∅	0	0	0	0
	1	1	1	1	1	1	1

→ dfs

n'

↓

①

boolean dfs(node, par) {

color[node] = 1;

for(int neighbor : adj[node]) {

if(color[neighbor] == 0)

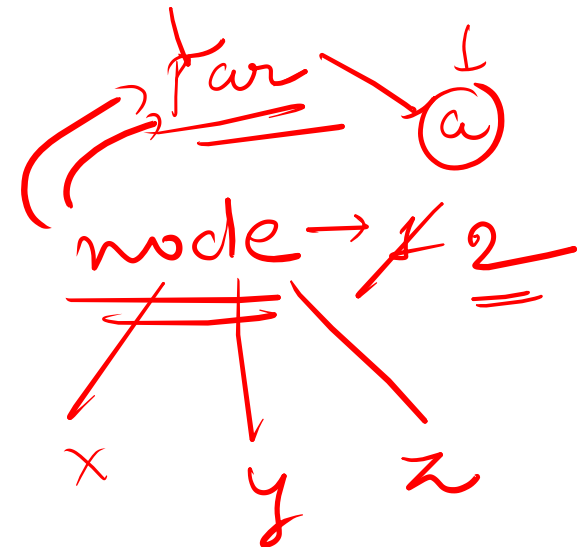
if(dfs(neighbor, node) == true)

return true;

else if(color[neighbor] == 1

neighbor != par) return true;

}



$O(V+E)$
↓
TC
 $O(V+E) \rightarrow \underline{\underline{SC}}$

```
color[node] = 2;  
return false;
```

Topological sort
directed graph

par $\rightarrow$ x

Print the path
that contains a
cycle
yourself //

